An AI-Based Document Translation System with Original Format Preservation

Hoang-Phuc Ha [1] Ngo-Ho Anh-Khoa [1] and Ngo-Ho Anh-Khoi [1*] | Hoang-Phuc Ha 1 Ngo-Ho Anh-khoi 1 and Ngo-Ho Anh-Khoi 1*

[1] Faculty of Information Technology, Nam Can Tho University, 168 Nguyen Van Cu Street, An Binh Ward, Can Tho City, Vietnam

phuc222683@studentnctu. edu. vn, nhakhoa@nctu. edu. vn, nhakhoi@nctu. edu. vn | 1 Faculty of Information Technology, Nam Can Tho University, 168 Nguyen Van Cu Street, An Binh Ward, Can Tho City, Vietnam

phuc222683@studentnctu. edu. vn, nhakhoa@nctu. edu. vn, nhakhoi@nctu. edu. vn

Abstract. In the context of the increasing demand for digital document translation, modern translation systems are expected not only to ensure the quality of translated content but also to preserve the original formatting of input documents. However, when translating structurally complex documents such as PDF and DOCX files, many systems still encounter problems such as layout distortion, table misalignment, image displacement, font errors, and limited support for bilingual display modes. This study proposes an artificial intelligence-based document translation system capable of processing two common document formats, DOCX and PDF, while supporting three translation modes: standard translation, side-by-side bilingual translation, and line-by-line bilingual translation. For DOCX documents, the system directly processes document components such as paragraphs, tables, headers, and footers. For PDF documents, the system applies a PDF→DOCX→translation→layout recovery→PDF pipeline to improve format preservation. The system integrates several translation providers, including Google Translate, DeepL Translator, and Gemini 2.5 Flash, to increase flexibility in translation quality and processing options. The experimental evaluation is conducted using two groups of criteria: formatting preservation and translation quality measured by BLEU Mean and BLEU Std. The results show that the proposed system achieves strong format preservation for DOCX documents and promising performance for academic PDF documents.

Keywords: Document translation, format preservation, artificial intelligence, PDF, DOCX, bilingual translation, BLEU.

1 Introduction

The rapid development of artificial intelligence and large language models has significantly advanced the field of natural language processing, especially machine translation. Modern translation models are no longer limited to converting text from a source language to a target language; they are also capable of handling context, terminology, and writing style more effectively than traditional statistical translation approaches(Bahdanau et al., 2015; Wu et al., 2016; Koehn, 2020). In particular, the Transformer architecture has become a fundamental foundation for many modern language processing models due to its ability to capture contextual relationships efficiently in long text sequences (Vaswani et al., 2017). In addition, the development of large language models such as GPT and unified text-to-text models has opened new directions for translation, summarization, and document processing tasks (Brown et al., 2020; Raffel et al., 2020). However, in practical scenarios, users do not only need to translate plain sentences or paragraphs. They often need to translate entire formatted documents such as scientific papers, technical reports, theses, contracts, and administrative forms. These documents are commonly stored in DOCX or PDF formats and contain various formatting components, including headings, paragraphs, tables, images, headers, footers, lists, and page layouts. Therefore, document translation should not be considered only as a content translation problem, but also as a formatting preservation problem.

In many existing document translation systems, preserving the original format remains a considerable challenge. DOCX has a relatively structured representation in which content is organized into paragraphs, tables, styles, headers, and footers, making it possible to process the document directly using programming tools. In contrast, PDF is mainly designed for display and printing, and its content is usually stored based on page coordinates rather than a logical document structure. As a result, when translated text becomes longer or shorter than the original text, problems such as text overflow, table misalignment, overlapping text, and layout changes may occur. Popular translation services such as Google Translate, DeepL, Adobe Acrobat, and AI platforms with file-upload support provide convenient solutions, but they still have limitations in terms of pipeline customization, bilingual presentation, and output format control. Based on these limitations, this study develops an AI-based document translation system that preserves the original document format and supports both DOCX and PDF files with three translation modes. The system uses different processing strategies for different document types: DOCX files are translated directly based on their structured paragraphs and tables, while PDF files are processed using a PDF→DOCX→translation→layout recovery→PDF pipeline. The proposed system is evaluated through two experimental groups: formatting preservation evaluation and translation quality evaluation, in order to assess both visual structure and linguistic content.

2 Related Work

In the field of document translation, many commercial systems have been developed to help users translate common file formats such as DOCX, PDF, and PPTX. Platforms such as DeepL Document Translation, Google Translate, Google Cloud Translation, and Adobe Acrobat provide online document translation features, allowing users to process multilingual documents for study, research, and professional work. These systems offer advantages in terms of processing speed, translation quality, and usability. However, most commercial platforms are closed systems, meaning that users cannot easily modify the internal processing pipeline, customize format preservation mechanisms, or flexibly control bilingual presentation modes such as side-by-side bilingual translation or line-by-line bilingual translation. For users working with academic or technical documents that contain complex structures, relying only on existing commercial file translation tools may not be sufficient to control formatting errors generated during the translation process.

From an academic perspective, machine translation has evolved significantly from statistical machine translation to neural machine translation and large language models. The attention mechanism in neural machine translation improved the ability to align source and target sentences(Bahdanau et al., 2015), while Google Neural Machine Translation demonstrated the effectiveness of neural networks in large-scale multilingual translation (Wu et al., 2016). Later, the Transformer architecture became a key foundation for modern translation and natural language processing models through its self-attention mechanism(Vaswani et al., 2017). Reviews of neural machine translation also indicate that translation quality has improved considerably, although challenges remain in context handling, domain-specific terminology, and evaluation methodology(Stahlberg, 2020; Koehn, 2020). In addition, large language models such as GPT and unified text-to-text models have expanded the ability to process long text, understand context, and generate more natural outputs (Brown et al., 2020; Raffel et al., 2020).

For document processing, DOCX and PDF formats have very different structural characteristics. DOCX is based on an XML structure, where content is organized into paragraphs, text runs, tables, styles, headers, and footers. This allows a system to iterate through document components, extract and replace text, and preserve most of the original formatting. In contrast, PDF is designed mainly for display and printing, and its text is often stored according to page coordinates. This makes reading-order detection, table recognition, and layout recovery more difficult. Studies on document layout analysis, such as PubLayNet, DocBank, FUNSD, DocVQA, and LayoutLM, show that understanding the visual and structural layout of documents is important for complex document processing tasks(Zhong et al., 2019; Li et al., 2020; Jaume et al., 2019; Mathew et al., 2021; Xu et al., 2020). In particular, LayoutLM and LayoutLMv2 combine textual, positional, and visual information to improve document understanding for visually rich documents (Xu et al., 2020; Xu et al., 2021).

A practical approach for PDF processing is to convert PDF files into an editable format such as DOCX, translate the extracted content, and then export the result back to PDF. The PDF→DOCX→translation→PDF pipeline has the advantage of using the flexible editing structure of Word documents, allowing the system to process text by paragraphs, tables, and styles instead of manipulating raw PDF coordinates directly. However, this pipeline also has limitations because the conversion process may introduce intermediate errors such as merged text, fragmented paragraphs, incorrectly detected tables, or modified page layouts. Therefore, for this pipeline to work effectively, preprocessing, DOCX sanitization, layout recovery, and output quality checking are required. For scanned documents or embedded images, OCR is also important for creating a text layer that can be processed, and Tesseract is one of the most widely used OCR engines in both research and practical applications (Smith, 2007).

Machine translation evaluation studies often use automatic metrics such as BLEU, TER, METEOR, and COMET to measure similarity between machine translations and reference translations(Papineni et al., 2002; Snover et al., 2006; Banerjee& Lavie, 2005; Rei et al., 2020). Among them, BLEU is a widely used metric based on n-gram overlap between a system-generated translation and a reference translation

(Papineni et al., 2002). However, BLEU scores must be reported clearly because they can vary depending on tokenization, normalization, and calculation settings(Post, 2018). For document translation with format preservation, translation quality alone is not sufficient. A system may translate content correctly but still fail in practical use if it loses tables, shifts images, distorts layout, or produces an unreadable output document. Therefore, this study evaluates the proposed system from two perspectives: formatting preservation and translation quality.

Based on the above analysis, the research gap lies in combining three factors: translation quality, original format preservation, and flexible bilingual presentation modes. Existing systems usually focus on either content translation or format preservation to a certain extent, but they do not fully support standard translation, side-by-side bilingual translation, and line-by-line bilingual translation within a customizable and extensible pipeline. Furthermore, allowing users to choose among translation providers such as Google Translate, DeepL, and Gemini makes the system more adaptable to different requirements in cost, speed, and translation quality. Therefore, this study proposes an AI-based document translation system that combines DOCX processing, a PDF→DOCX→translation→PDF pipeline, special-content protection, layout recovery, and experimental evaluation of both formatting and translation quality.

3 System Design

The proposed system is designed to solve the problem of translating documents while preserving the original format for two main document types: DOCX and PDF. The overall workflow begins when a user uploads a document through the web interface and selects the target language, translation provider, and translation mode. After receiving the request, the system validates the file format, stores the input document, and creates a background processing job. This asynchronous processing design is suitable for large documents or documents that require multiple intermediate processing steps, especially PDF files. During translation, the user interface can monitor the processing status through a job identifier and allow the user to download the output file when the job is completed.

The system architecture consists of several main components: the user interface, the request-handling API, the background job manager, the file processing service, the translation provider module, the layout recovery module, and the output generation module. The user interface allows users to upload DOCX or PDF files, select the target language, choose a translation provider such as Google Translate, DeepL, or Gemini, and select one of three translation modes: standard translation, side-by-side bilingual translation, and line-by-line bilingual translation. The API validates the uploaded file, stores it, creates a translation record, and initializes a background job. The file processing service then determines the document type and selects the appropriate pipeline. This architecture allows the system to be extended with additional translation providers or new document formats in the future.

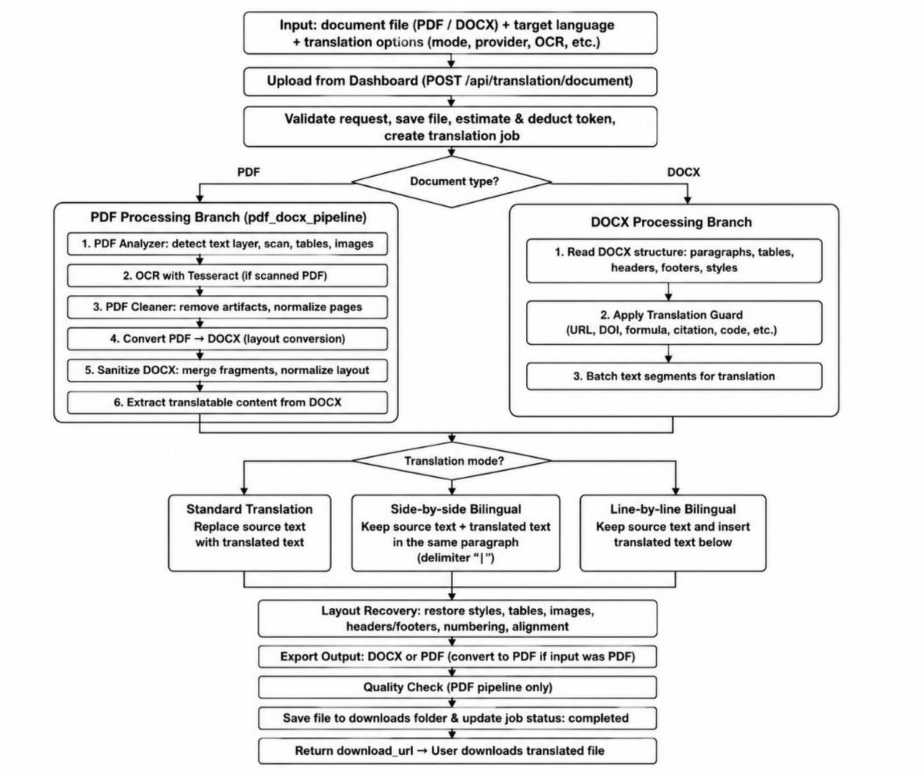

Fig. 1.*Workflow of the proposed AI-based document translation system with original format preservation.*

For DOCX documents, the system directly processes the document structure by iterating through paragraphs, tables, headers, and footers. Since DOCX has a clear internal structure, the system can preserve most styles, fonts, margins, and table structures while modifying only textual content. Before sending text to a translation provider, the system applies a Translation Guard mechanism to protect non-translatable elements such as URLs, DOIs, email addresses, citations, formulas, technical symbols, and code snippets. This mechanism is especially important for academic and technical documents because translating such elements incorrectly may change their original meaning. After translation, the translated text is written back into the document according to the selected translation mode.

For PDF documents, the system uses an intermediate conversion pipeline: PDF→DOCX→translation→layout recovery→PDF. First, the PDF file is analyzed to determine whether it contains a text layer, whether it is a scanned document, and whether it includes tables, images, or complex page layouts. If the file is scanned or contains image-based text, OCR can be applied to create a processable text layer. The PDF is then cleaned and converted into DOCX. Since PDF-to-DOCX conversion may create issues such as fragmented paragraphs, merged text, incorrect table detection, or abnormal line spacing, the system performs a sanitization step to merge fragmented segments, normalize the layout, and reduce noise before translation. After translation, the Layout Recovery module is used to restore the document layout before exporting the result back to PDF.

The system supports three translation modes to meet different user needs. In standard translation mode, the original text is replaced by the translated text, which is suitable when the user only needs the final document in the target language. In side-by-side bilingual translation mode, the system keeps the original text and places the translated text in the same paragraph, usually separated by the"|" delimiter, allowing users to quickly compare the source and target text. In line-by-line bilingual translation mode, the system keeps the original paragraph and inserts the translated paragraph immediately below it. This mode is useful for language learning, academic reading, and translation checking. Since bilingual modes increase the length of the document, the goal of layout recovery is not to preserve the exact number of pages, but to ensure that the content remains readable, correctly ordered, complete, and free from serious text-overlapping errors.

```
Algorithm 1
Input:
    F: input document (. pdf or . docx)
    L: target language
    P: translation provider (Google Translate, DeepL, Gemini)
    M: translation mode (standard, side-by-side bilingual, line-by-line bilingual)
    Opt: optional settings such as OCR and delimiter configuration | Algorithm
1
Input:
    F: input document (. pdf or . docx)
    L: target language
    P: translation provider (Google Translate, DeepL, Gemini)
    M: translation mode (standard, side-by-side bilingual, line-by-line bilingual)
Opt: optional settings such as OCR and delimiter configuration

Output:
    F_out: translated document with preserved formatting

BEGIN
    Validate the input document F and target language L

    IF the file extension of F is not . pdf or . docx THEN
        Return unsupported file format error
    END IF | IF the file extension of F is not . pdf or .
    docx THEN              Return unsupported file format
    error
    END IF

    Save F to upload storage
    Create a background translation job
    Return job_id and status_url to the client

    Set the job status to in_progress

    IF F is a PDF document THEN
        Analyze the PDF structure, including text layer, scanned pages, tables, and
images

        IF the PDF is scanned THEN
            Apply OCR to create a searchable PDF
```

IF OCR fails in strict mode THEN

```
            Set the job status to failed
            Return error
          END IF
        END IF

        Clean the PDF document
        Convert the PDF document to DOCX
        Sanitize the converted DOCX document

        Extract paragraphs, tables, headers, and footers from the DOCX document
        Apply translation guard to protect URLs, DOIs, citations, formulas, and
special symbols
        Translate the extracted content using provider P and mode M
        Recover the document layout after translation | Extract paragraphs, tables,
headers, and footers from the DOCX document     Apply translation guard to
protect URLs, DOIs, citations, formulas, and   special symbols
        Translate the extracted content using provider P and mode M
        Recover the document layout after translation

        IF M is standard translation THEN
          Apply the final layout refinement step
        END IF

        Convert the translated DOCX document back to PDF
        Perform quality checking on the output PDF

    ELSE IF F is a DOCX document THEN
        Open the DOCX document directly
        Extract paragraphs, tables, headers, and footers
        Apply translation guard to protect special content
        Translate the DOCX content using provider P and mode M
        Preserve the original styles and document structure
    END IF | ELSE IF F is a DOCX document THEN
        Open the DOCX document directly
        Extract paragraphs, tables, headers, and footers
        Apply translation guard to protect special content
        Translate the DOCX content using provider P and mode M
        Preserve the original styles and document structure
    END IF

    Save the translated document to download storage
    Set the job status to completed
    Return the download path of F_out

    IF any error occurs THEN
        Set the job status to failed with the error message
    END IF

    The client periodically checks status_url
    IF the job status is completed THEN
    Download the translated document
    END IF
END | The client periodically checks
status_url       IF the job status is
```

```
completed THEN              Download
the translated document     END IF
END
```

The pseudocode describes the overall processing workflow of the proposed system. For DOCX documents, the system directly processes paragraphs, tables, headers, and footers to translate textual content while preserving the original formatting. For PDF documents, the system uses DOCX as an intermediate editable format by converting the PDF into DOCX, translating the extracted content, recovering the layout, and exporting the result back to PDF. During translation, the system supports three modes: standard translation, side-by-side bilingual translation, and line-by-line bilingual translation. In addition, special elements such as URLs, DOIs, citations, formulas, and technical symbols are protected to prevent incorrect translation of non-translatable content.

After translation and layout recovery are completed, the system exports the output file according to the original document type. If the input file is DOCX, the system returns a translated DOCX file. If the input file is PDF, the system converts the translated DOCX back to PDF and performs output quality checking. This checking step aims to detect serious errors such as missing content, missing tables, missing images, or layout problems that affect readability. The final file is saved in the download directory, and the download path is returned to the user after the job status changes to completed.

4 Experiments

The experiments were conducted to evaluate the system from two main perspectives: formatting preservation after translation and translation content quality. Separating these two evaluation groups is necessary because document translation depends not only on linguistic quality but also on the ability to preserve the original visual structure of the document. A translation may be semantically correct but still fail in practice if it loses tables, changes the layout, or produces an unreadable output file. Conversely, a document may preserve its formatting well, but the translation may still be insufficient if it is unnatural or uses incorrect terminology.

The experiments were conducted on an Acer Nitro AN515-45 computer running Microsoft Windows 11 Home Single Language 64-bit. The machine uses an AMD Ryzen 55600H with Radeon Graphics processor, with 12 logical processors at approximately 3.3 GHz, on an x64-based PC architecture. The system is equipped with 8 GB of RAM and DirectX 12. The development environment uses Python together with the FastAPI framework on the backend for receiving document translation requests, managing background processing jobs, and returning output files to users. The user interface is implemented using HTML, CSS, and JavaScript. For document processing, the system uses pdf2docx for PDF-to-DOCX conversion, DOCX processing libraries for handling paragraphs, tables, headers, and footers, Tesseract OCR for scanned documents or embedded images, and a DOCX-to-PDF conversion tool for generating the final translated PDF output. The translation providers used in the experiments include Google Translate, DeepL Translator, and Gemini 2.5 Flash.

The experimental data consists of DOCX and PDF academic documents containing headings, paragraphs, tables, images, and page layouts. These documents were processed using three translation modes: standard translation, side-by-side bilingual translation, and line-by-line bilingual translation. For format evaluation, the system considers the following criteria: Heading, Text/Style, Table, Layout, and No

Overflow. Heading measures whether titles and section hierarchy are preserved. Text/Style evaluates whether fonts, paragraphs, and textual presentation remain readable. Table measures whether table structures are preserved. Layout reflects the overall page organization. No Overflow measures whether the output document avoids serious text overflow or overlapping. Minor errors that can be manually corrected, such as slight line shifts or natural page increases caused by longer translated text, are not considered serious errors if they do not affect readability.

The AVG score is calculated as the arithmetic mean of five formatting evaluation criteria: Heading, Text/Style, Table, Layout, and No Overflow. A higher AVG value indicates better formatting preservation and output readability.

relative comparison among machine translation outputs rather than an absolute conclusion about translation quality.

Overall, the experimental results show that the proposed system performs well for DOCX documents and achieves promising results for PDF documents. The difference between DOCX and PDF performance comes from the structural characteristics of each format. DOCX allows direct processing of text and formatting components, while PDF requires several intermediate conversion steps, which increases the possibility of formatting errors. For bilingual translation modes, document length and page count may increase because both the source and translated text are retained. This should not be considered a formatting error as long as the document remains readable, complete, and free from serious text overlapping.

Despite the obtained results, the system still has several limitations that need further improvement. The PDF→DOCX→translation→PDF pipeline cannot guarantee perfect formatting for all types of PDF documents, especially documents with multi-column layouts, complex tables, mathematical formulas, or professional desktop-publishing designs. In addition, scanned PDFs depend heavily on OCR quality; if images are blurred, skewed, noisy, or use unusual fonts, OCR errors may affect translation quality. Since BLEU in this study is used in a cross-output setting due to the absence of a complete manually reviewed reference translation dataset, future work should construct a reference dataset and combine additional metrics such as METEOR, TER, and COMET for a more comprehensive evaluation(Banerjee& Lavie, 2005; Snover et al., 2006; Rei et al., 2020).

5 Conclusion

This study proposed and developed an AI-based document translation system with original format preservation for two main document formats: DOCX and PDF. The system supports three translation modes: standard translation, side-by-side bilingual translation, and line-by-line bilingual translation. It also allows users to select different translation providers, including Google Translate, DeepL Translator, and Gemini 2.5 Flash. For DOCX documents, the system directly processes structured components such as paragraphs, tables, headers, and footers. For PDF documents, the system uses a PDF→DOCX→translation→layout recovery→PDF pipeline to take advantage of DOCX editability and improve format preservation.

The experimental results show that the system achieves very strong format preservation for Word documents and promising results for PDF documents. The bilingual translation modes are useful for helping users compare the original content with the translated content, especially in learning, research, and terminology checking. In terms of translation quality, BLEU Mean and BLEU Std show that the translation providers achieved relatively close results on the test dataset, with Gemini 2.5 Flash obtaining the highest score in the cross-output BLEU experiment. However, to evaluate translation quality more accurately, future research should construct manually reviewed reference translations and combine additional evaluation metrics.

In future work, the system can be improved in several directions. First, PDF processing can be enhanced by improving layout analysis, table recognition, and post-translation layout recovery. Second, OCR quality should be improved for scanned | In future work, the system can be improved in several directions. First, PDF processing can be enhanced by improving layout analysis, table recognition, and post-translation layout recovery. Second, OCR quality should be improved for scanned

PDFs and embedded images. Third, a domain-specific glossary can be added to ensure terminology consistency in academic and technical documents. Finally, support for additional formats such as PPTX, XLSX, and HTML can be developed to broaden the practical applicability of the system.

References

1. Bahdanau, D., Cho, K.,&Bengio, Y.(2015). *Neural machine translation by jointly learning to align and translate*International Conference on Learning Representations. https://arxiv. org/pdf/1409.0473
2. Banerjee, S., &Lavie, A.(2005). *METEOR: An automatic metric for MT evaluation with improved correlation with human judgments*Proceedings of the ACL Workshop on Intrinsic and Extrinsic Evaluation Measures for Machine Translation and/or | Intrinsic and Extrinsic Evaluation Measures for Machine Translation and/or Summarization, 65–72. https://aclanthology. org/W05-0909. pdf
3. Brown, T. B., Mann, B., Ryder, N., Subbiah, M., Kaplan, J., Dhariwal, P., Neelakantan, A., Shyam, P., Sastry, G., Askell, A., Agarwal, S., Herbert-Voss, A., Krueger, G., Henighan, T., Child, R., Ramesh, A., Ziegler, D. M., Wu, J., Winter, C.,Amodei, D.(2020). *Language models are few-shot learners*Advances in Neural Information Processing Systems, 33, 1877–1901. https://arxiv. org/pdf/2005.14165
4. Costa-jussà, M. R., Cross, J., Çelebi, O., Elbayad, M., Heafield, K., Heffernan, K., | 4. Costa-jussà, M. R., Cross, J., Çelebi, O., Elbayad, M., Heafield, K., Heffernan, K., Kalbassi, E., Lam, J., Licht, D., Maillard, J., Sun, A., Wang, S., Wenzek, G., Youngblood, A., Akula, B., Barrault, L., Gonzalez, G. M., Hansanti, P., Hoffman, J., . Wang, J. (2022). | A., Akula, B., Barrault, L., Gonzalez, G. M., Hansanti, P., Hoffman, J., . Wang, J. (2022). *No language left behind: Scaling human-centered machine translation*arXiv preprint arXiv: 2207.04672. https://arxiv. org/pdf/2207.04672
5. Devlin, J., Chang, M.-W., Lee, K.,&Toutanova, K.(2019). *BERT: Pre-training of deep bidirectional transformers for language understanding*Proceedings of NAACL-HLT, 4171–4186. https://aclanthology. org/N19-1423. pdf
6. Jaume, G., Ekenel, H. K.,&Thiran, J.-P.(2019). *FUNSD: A dataset for form understanding in noisy scanned documents*International Conference on Document Analysis and Recognition Workshops. https://arxiv. org/pdf/1905.13538
7. Junczys-Dowmunt, M., Grundkiewicz, R., Dwojak, T., Hoang, H., Heafield, K., | 7. Junczys-Dowmunt, M., Grundkiewicz, R., Dwojak, T., Hoang, H., Heafield, K., Neckermann, T., Seide, F., Germann, U., Aji, A. F., Bogoychev, N., Martins, A. F. T.,& | Neckermann, T., Seide, F., Hermann, U., Aji, A. F., Bogoychev, N., Martins, A. F. T.,& Birch, A.(2018). *Marian: Fast neural machine translation in C++*Proceedings of ACL System Demonstrations, 116–121. https://aclanthology. org/P18-4020. pdf
8. Koehn, P. (2020). *Neural machine translation* Cambridge University Press. https://arxiv. org/pdf/1709.07809
9. Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, | 9. Lewis, P., Perez, E., Pictus, A., Petroni, F., Karpukhin, V., Goyal, N., Kuettler, H., Lewis, M., Yih, W., Rocktäschel, T., Riedel, S.,&Kiela, D.(2020). *Retrieval-augmented* | M., Yih, W., Rocktäschel, T., Riedel, S.,&Kiela, D.(2020). Retrieval-augmented *generation for knowledge-intensive NLP tasks*Advances in Neural Information Processing Systems, 33, 9459–9474. https://arxiv. org/pdf/2005.11401
10. Li, M., Xu, Y., Cui, L., Huang, S., Wei, F., Li, Z.,&Zhou, M.(2020). *DocBank: A benchmark dataset for document layout analysis*Proceedings of the 28th International Conference on Computational Linguistics, 949–960. https://aclanthology. org/2020. coling-main.82. pdf
11. Lin, C.-Y.(2004). *ROUGE: A package for automatic evaluation of summaries*

Proceedings of the Workshop on Text Summarization Branches Out, 74–81. https://aclanthology. org/W04-1013. pdf

12. Mathew, M., Karatzas, D.,&Jawahar, C. V.(2021). *DocVQA: A dataset for VQA on document images*Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision, 2200–2209. https://openaccess. thecvf. com/content/WACV2021/papers/Mathew_DocVQA_A_Dataset _for_VQA_on_Document_Imagespaper. pdf
13. Papineni, K., Roukos, S., Ward, T.,&Zhu, W.-J.(2002). BLEU: A method for automatic evaluation of machine translationProceedings of the 40th Annual Meeting of the Association for Computational Linguistics, 311–318. https://aclanthology. org/P02-1040. pdf
14. Post, M.(2018). A call for clarity in reporting BLEU scoresProceedings of the Third Conference on Machine Translation, 186–191. https://aclanthology. org/W18-6319. pdf
15. Raffel, C., Shazeer, N., Roberts, A., Lee, K., Narang, S., Matena, M., Zhou, Y., Li, W., & Liu, P. J.(2020). Exploring the limits of transfer learning with a unified text-to-text transformer Journal of Machine Learning Research, 21(140), 1–67. https://jmlr. org/papers/volume21/20-074/20-074. pdf
16. Rei, R., Stewart, C., Farinha, A. C., & Lavie, A. (2020). COMET: A neural framework for machine translation evaluationProceedings of the 2020 Conference on Empirical Methods in Natural Language Processing, 2685–2702. https://aclanthology. org/2020. emnlp-main.213. pdf
17. Smith, R.(2007). An overview of the Tesseract OCR engineProceedings of the International Conference on Document Analysis and Recognition, 629–633. https://research. google. com/pubs/archive/33418. pdf
18. Snover, M., Dorr, B., Schwartz, R., Micciulla, L.,&Makhoul, J.(2006). A study of translation edit rate with targeted human annotationProceedings of the Association for Machine Translation in the Americas, 223–231. https://aclanthology. org/2006. amta-papers.25. pdf
19. Stahlberg, F.(2020). Neural machine translation: A reviewJournal of Artificial Intelligence Research, 69, 343–418. https://arxiv. org/pdf/1912.02047
20. Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I.(2017). Attention is all you needAdvances in Neural Information Processing Systems, 30. https://arxiv. org/pdf/1706.03762
21. Wu, Y., Schuster, M., Chen, Z., Le, Q. V., Norouzi, M., Macherey, W., Krikun, M., Cao, Y., Gao, Q., Macherey, K., Klingner, J., Shah, A., Johnson, M., Liu, X., Kaiser, Ł., Gouws, S., Kato, Y., Kudo, T., Kazawa, H., Dean, J.(2016). Google's neural machine translation system: Bridging the gap between human and machine translationarXiv preprint arXiv: 1609.08144. https://arxiv. org/pdf/1609.08144
22. Xu, Y., Li, M., Cui, L., Huang, S., Wei, F., & Zhou, M. (2020). LayoutLM: Pre-training of text and layout for document image understandingProceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 1192–1200. https://arxiv. org/pdf/1912.13318
23. Xu, Y., Xu, Y., Lv, T., Cui, L., Wei, F., Wang, G., Lu, Y., Florencio, D., Zhang, C., Che, W., Zhang, M., Zhou, L., Huang, Q.,&Zhou, M. (2021). LayoutLMv2: Multi-modal pre-training for visually-rich document understandingProceedings of the Annual Meeting of the Association for Computational Linguistics. https://aclanthology. org/2021. acl-long.201. pdf
24. Zhong, X., Tang, J., & Yepes, A. J. (2019). PubLayNet: Largest dataset ever for document layout analysisProceedings of the International Conference on Document Analysis and Recognition, 1015–1022. https://arxiv. org/pdf/1908.07836